

QUANTUM MECHANICS

SECTION

Why do we need
quantum mechanics?

LECTURE

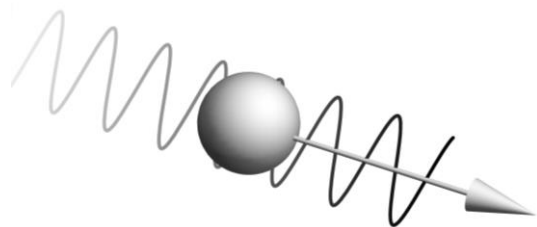
What is light?
A wave?

What is light?

“The whole 50 years of conscious brooding did not bring me closer to the answer to the question 'What are light quanta'. Today every scoundrel thinks he knows, but he is wrong ...”
- Einstein to Besso, 1951

Electrons

- Particle
 - Mass $m = 9.109 \times 10^{-31}$ kg
 - Charge $q = -e = -1.602 \times 10^{-19}$ As
 - Radius $r = 2.817 \times 10^{-15}$ m



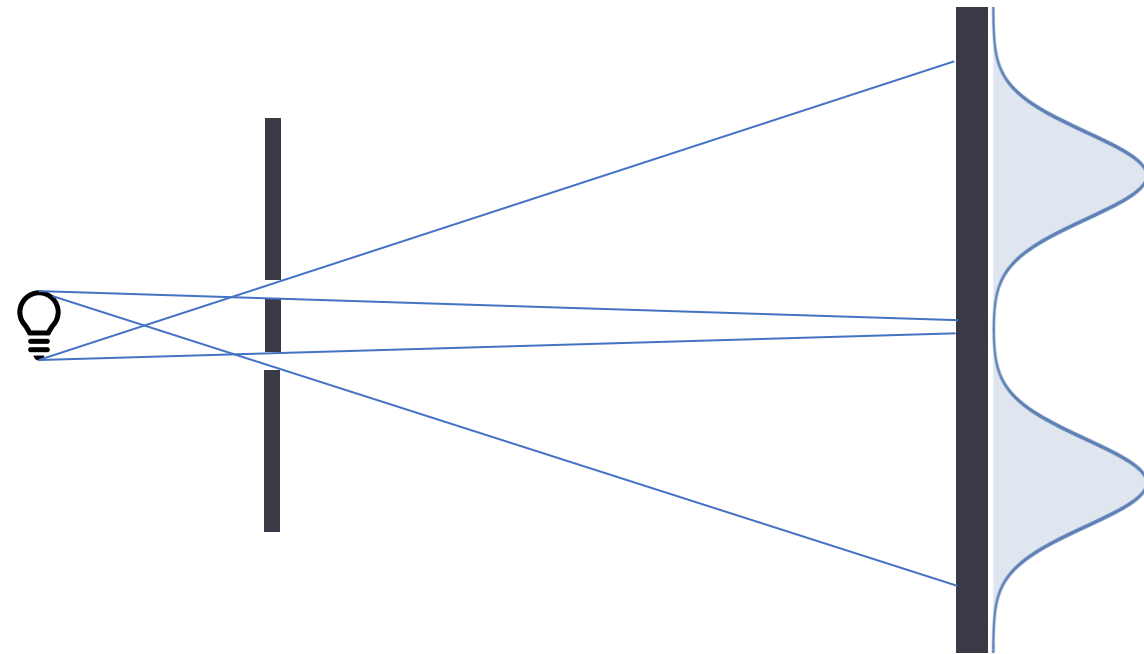
Light

- Particle
 - No mass $m = 0$ kg
 - No charge $q = 0$ As
 - 1899-1915: Photoelectric effect
 - 1900: Planck law for black-body radiation
- Wave
 - 1864: Maxwell's electrodynamics
 - 1802: Double-slit experiment

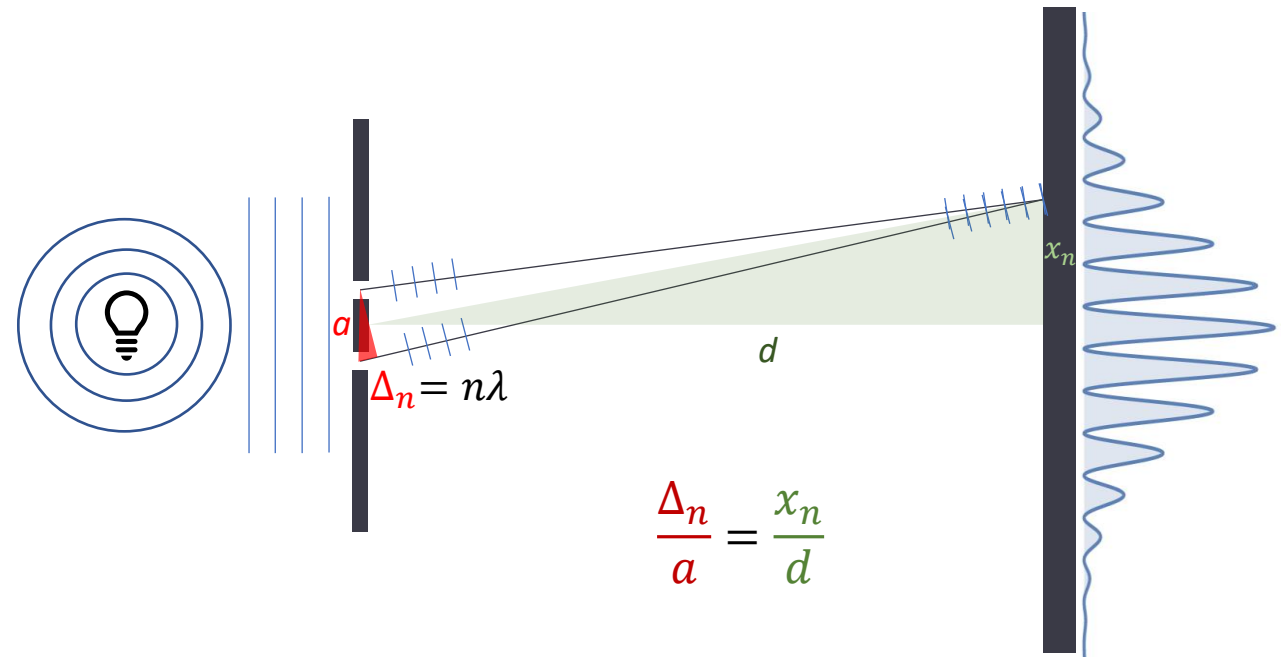
1924: Wave-particle dualism

Double-slit experiment

Light as particle



Light as wave



Intensity maxima distribution in the experiment

$$x_n = \frac{d}{a} \lambda n$$

Light as a wave

$$\frac{\partial}{\partial x} f(x) = f'(x)$$

Laplace operator

$$\Delta = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

Maxwell's equations

$$\nabla \cdot \mathbf{E} = 0$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\nabla \cdot \mathbf{B} = 0$$

$$\nabla \times \mathbf{B} = \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}$$

$$\nabla \times (\nabla \times \mathbf{E}) = \nabla \times \left(-\frac{\partial \mathbf{B}}{\partial t} \right) = -\frac{\partial}{\partial t} (\nabla \times \mathbf{B}) = -\mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2} = -\frac{1}{c^2} \frac{\partial^2 \mathbf{E}}{\partial t^2}$$

$$\nabla \times (\nabla \times \mathbf{E}) = \nabla(\nabla \cdot \mathbf{E}) - \Delta \mathbf{E}$$

$$\nabla \cdot \mathbf{E} = 0$$

Wave equation

$$\frac{1}{c^2} \frac{\partial^2 \mathbf{E}}{\partial t^2} - \Delta \mathbf{E} = 0$$

$$\mathbf{E}(\mathbf{r}, t) = E_0 \cos(\omega t - \mathbf{k} \cdot \mathbf{r})$$

Light as an electromagnetic wave

$$\mathbf{E}(\mathbf{r}, t) = E_0 \operatorname{Re}\{e^{i(\omega t - \mathbf{k} \cdot \mathbf{r})}\}$$

QUANTUM MECHANICS

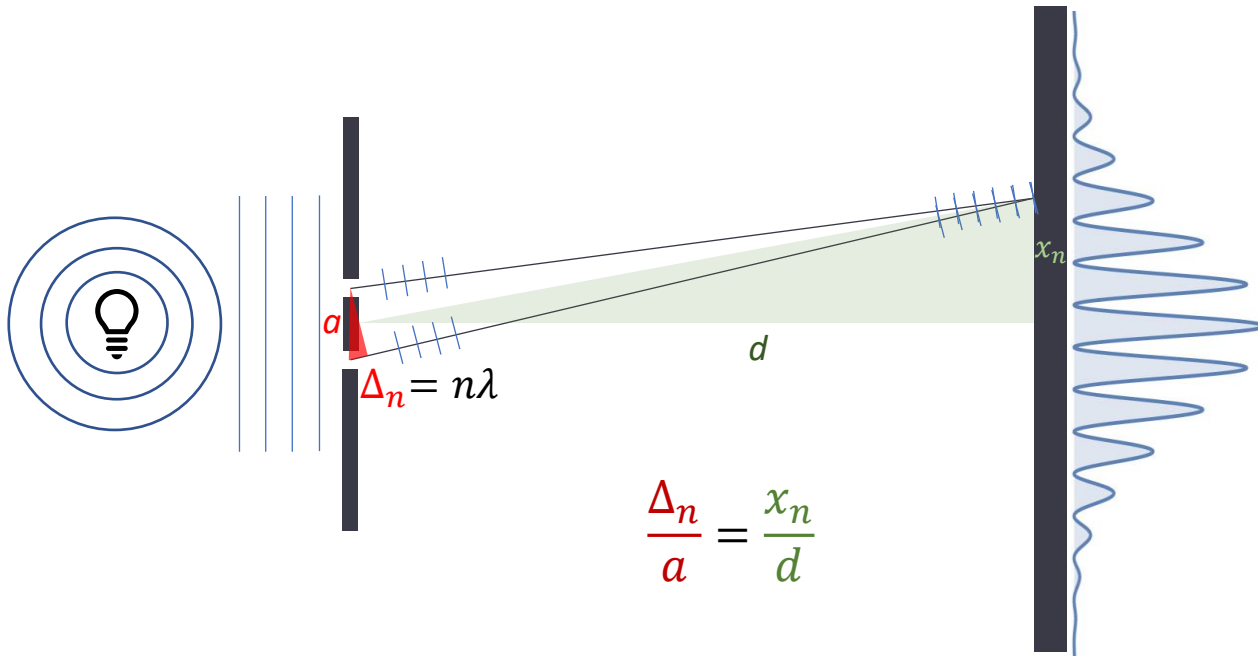
SECTION

Why do we need
quantum mechanics?

LECTURE

What is light?
A particle?

Light as a wave



- Wave equation

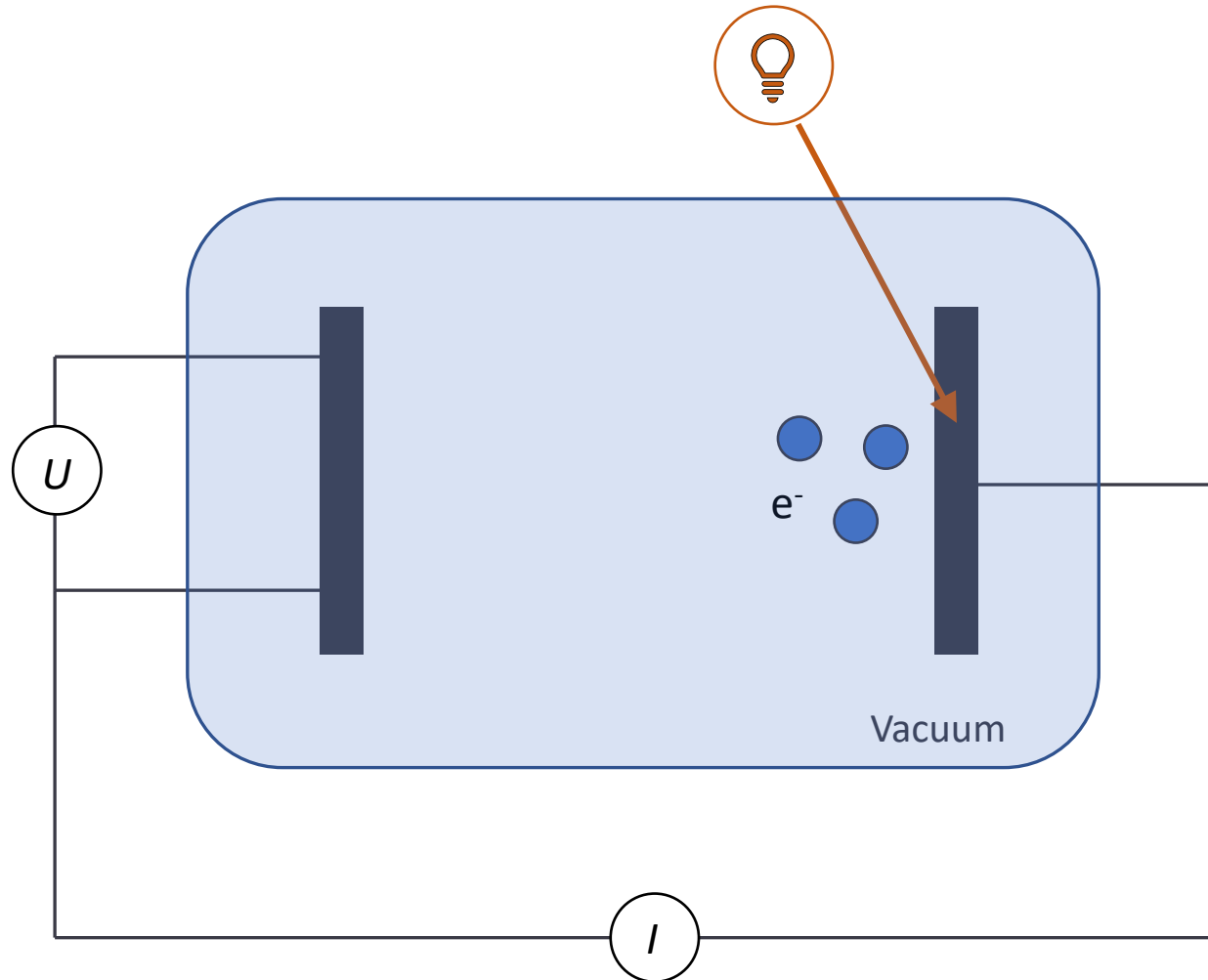
$$\frac{1}{c^2} \frac{\partial^2 E}{\partial t^2} - \Delta E = 0$$

- Light as an electromagnetic wave

$$E(\mathbf{r}, t) = E_0 \cos(\omega t - \mathbf{k} \cdot \mathbf{r})$$

- Energy, Intensity proportional to the amplitude E_0

Photoelectric effect



- Electrons exit the metal
 - Heating?
 - Takes some time?
- Move towards the other contact due to their kinetic energy

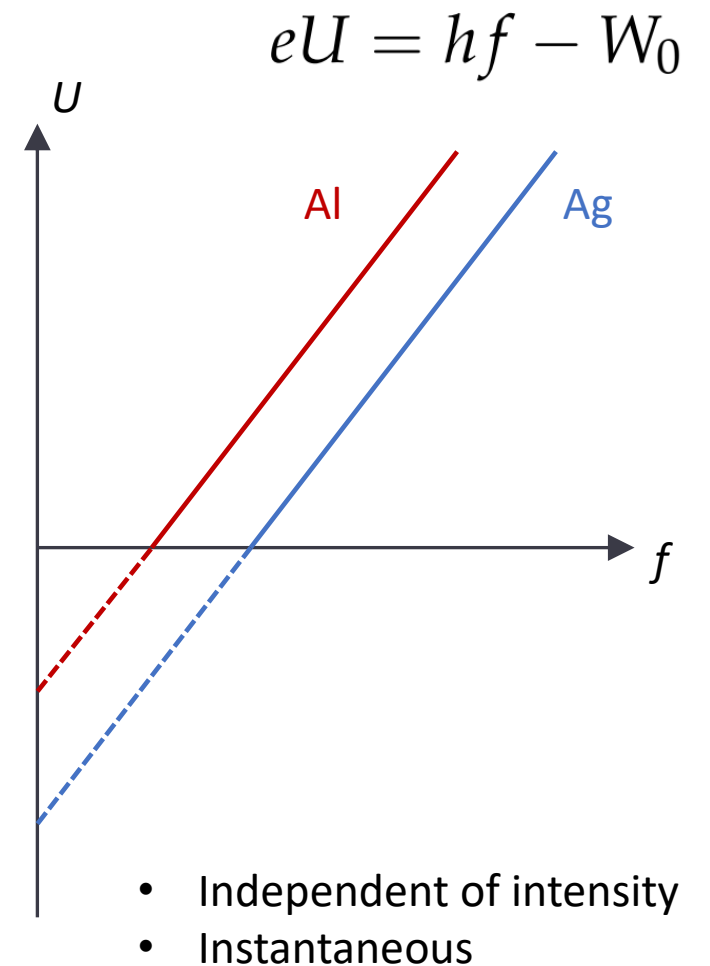
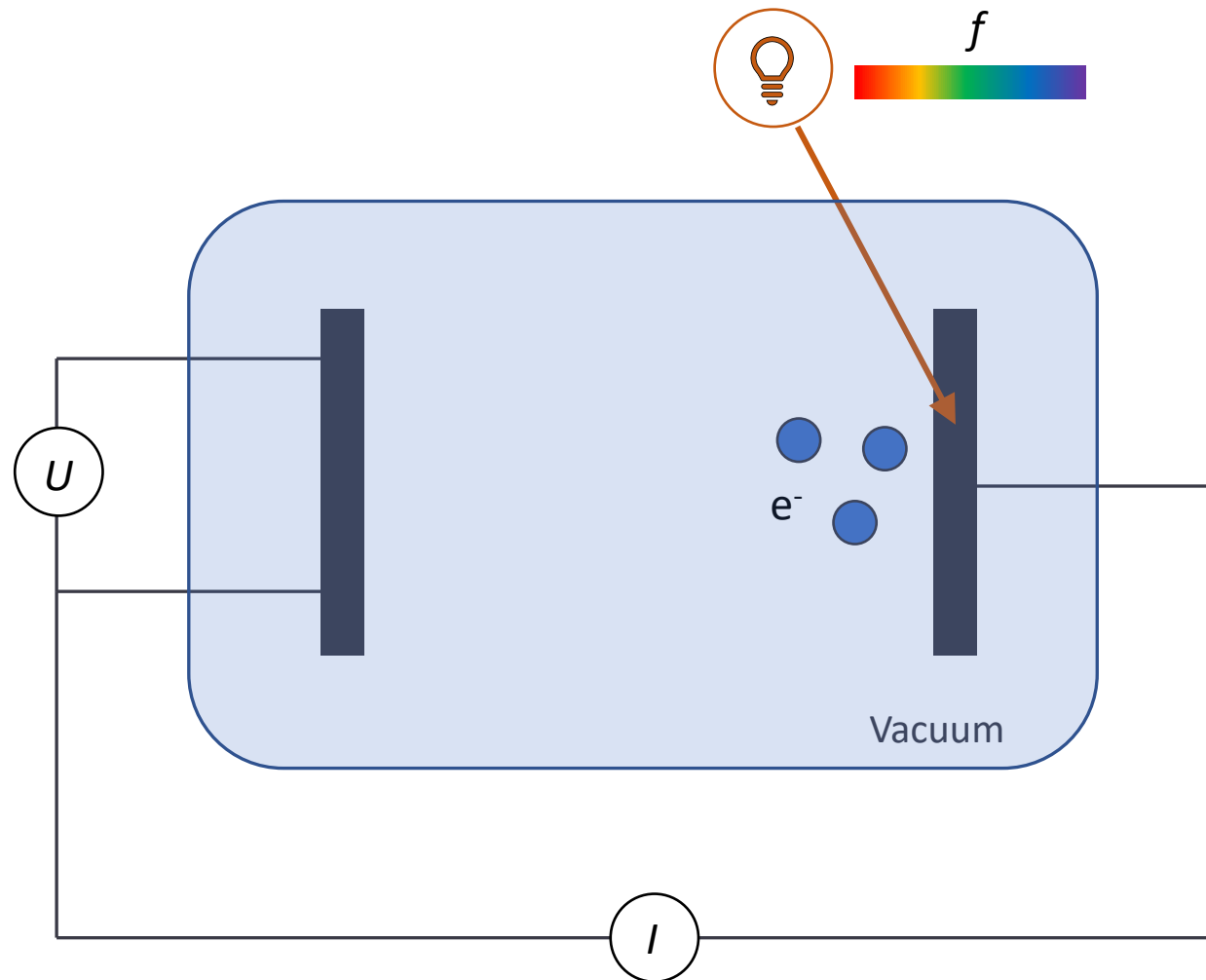
$$E_{\text{kin}} = \frac{m}{2}v^2 = \frac{p^2}{2m}$$

- Voltage U leads to a counteracting field
 - Electrons stop ($I = 0$) when

$$E_{\text{kin}} = eU$$

- We can measure E_{kin} indirectly

Photoelectric effect



Photoelectric effect

$$eU = hf - W_0$$

$$hf = E_{\text{kin}} + W_0$$

Wave

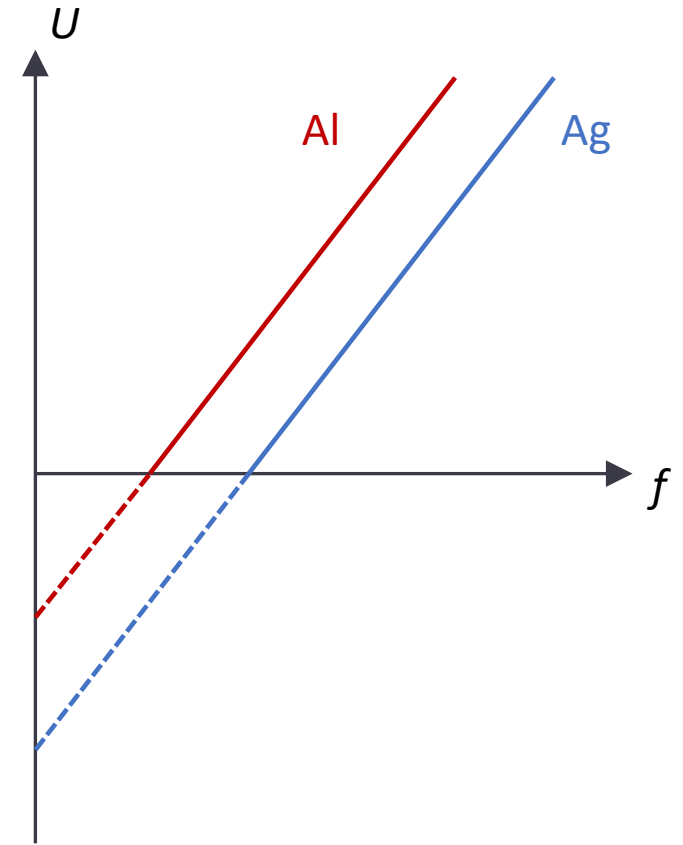
$$E(\mathbf{r}, t) = E_0 \cos(\omega t - \mathbf{k} \cdot \mathbf{r})$$

- Independent of intensity
- Instantaneous
- Only happens above critical frequency

→ Transmitted energy of the light comes in portions

→ Light has particle-like properties (quantum)

$$E_{\text{photon}} = hf = \hbar\omega$$



QUANTUM MECHANICS

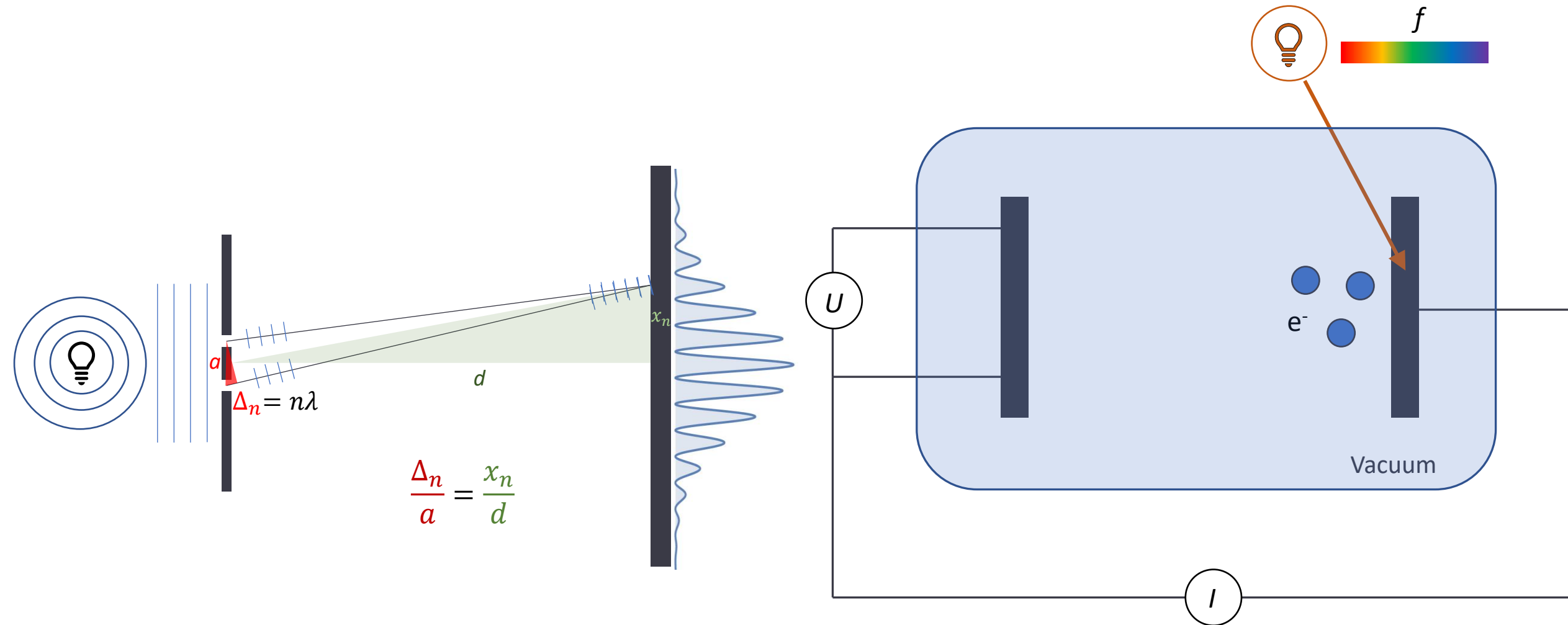
SECTION

Why do we need
quantum mechanics?

LECTURE

Particle-wave dualism
of matter

Light as wave and particle



Wave-particle dualism

- 1905 Einstein: particle-wave dualism for light
- 1924 De Broglie: particle-wave dualism for classical particles have wave-like properties

Light

$$E = \hbar\omega = \frac{hc}{\lambda}$$

$$E^2 - p^2c^2 = m^2c^4 \xrightarrow{m=0} E = pc$$

$$\lambda = \frac{h}{p}$$

Conventional particles

$$p = mv$$

De Broglie wave length

$$\lambda = \frac{h}{mv}$$

- Human (80 kg walking at 10 km/h): $\lambda = 3 \times 10^{-36} \text{ m}$
- Electron in E-field (U = 10 V): $\lambda = 9 \times 10^{-12} \text{ m}$

What is ~~light~~ an electron?

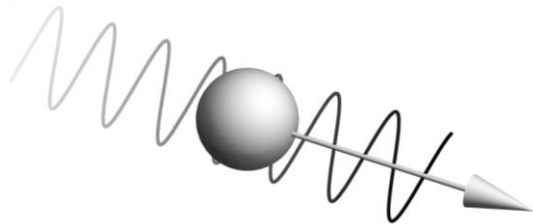
“The whole 50 years of conscious brooding did not bring me closer to the answer to the question 'What are light quanta'. Today every scoundrel thinks he knows, but he is wrong ...”
- Einstein to Besso, 1951

Electrons

- Particle
 - Mass $m = 9.109 \times 10^{-31}$ kg
 - Charge $q = -e = -1.602 \times 10^{-19}$ As
- Wave
 - 1961: Double-slit experiment

Light

- Particle
 - 1899-1915: Photoelectric effect
 - 1900: Planck law for black-body radiation
- Wave
 - 1802: Double-slit experiment



1924: Wave-particle dualism

QUANTUM MECHANICS

SECTION

Why do we need
quantum mechanics?

NEXT LECTURE

Heisenberg's uncertainty
& Schrödinger's cat

Heisenberg uncertainty

- 1927 Heisenberg:

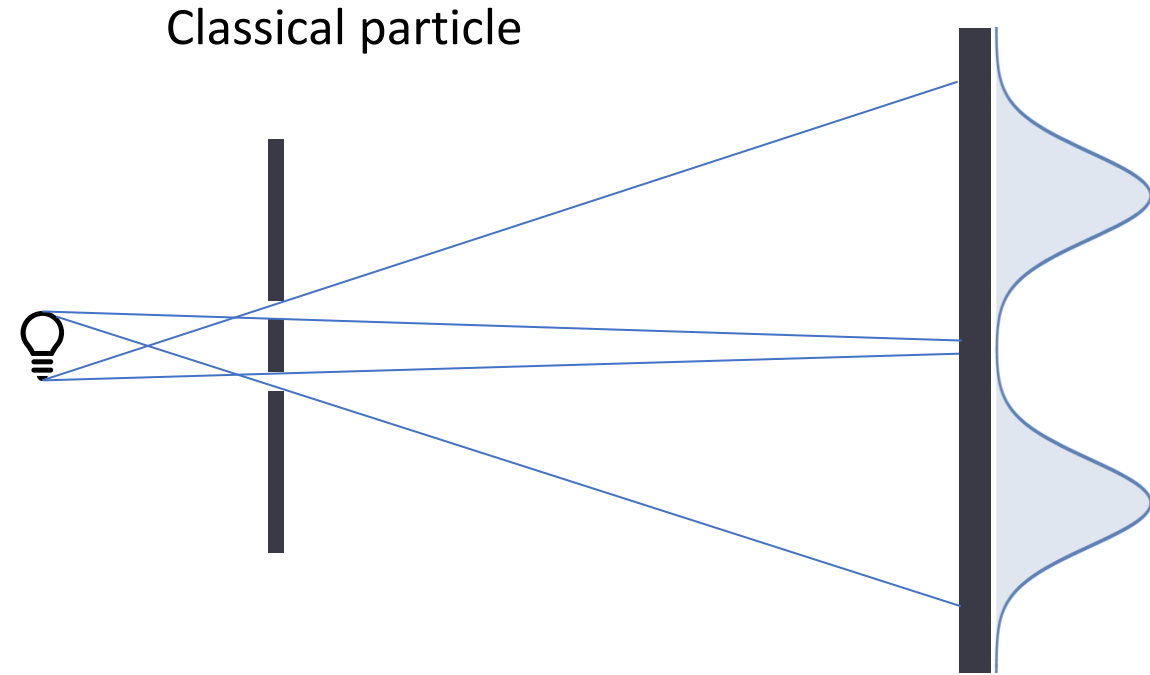
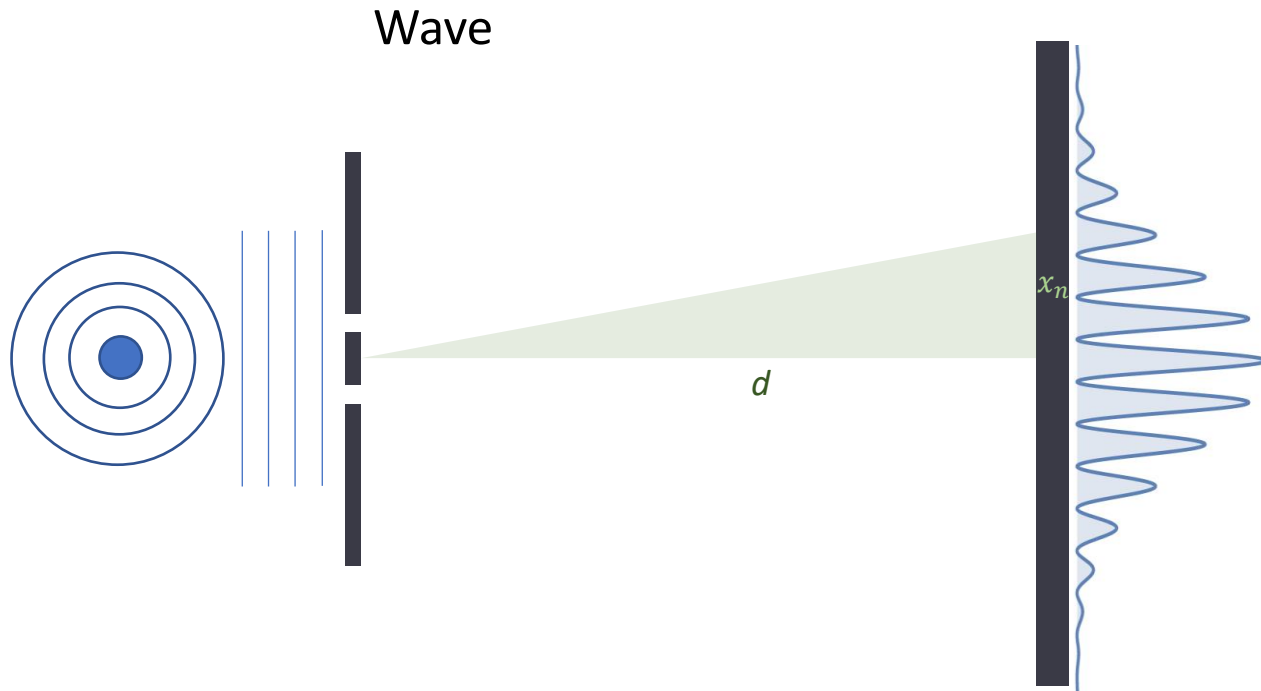
Position and momentum cannot be measured accurately at the same time

- If we measure an electron (e.g. we have to apply light), at least one photon will affect the electron and will change its momentum

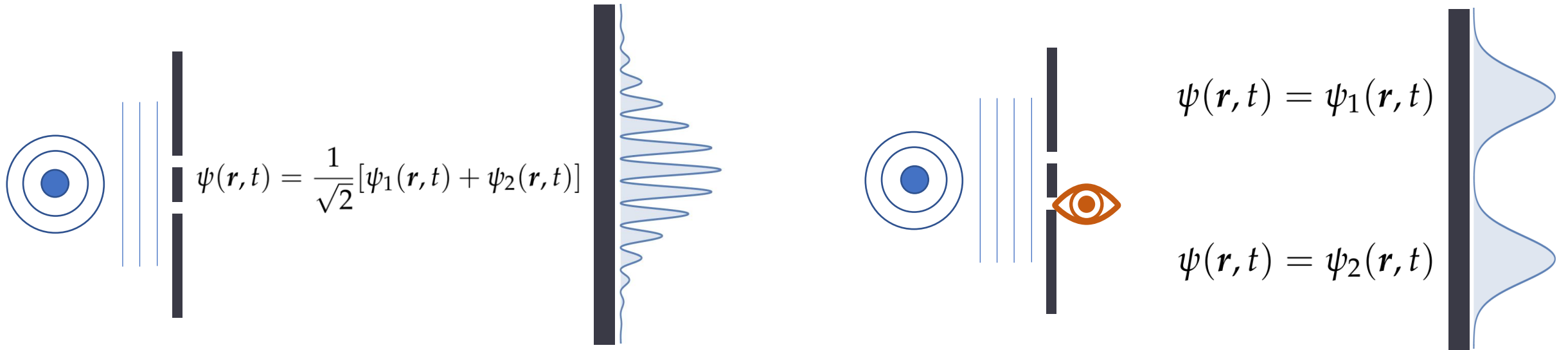
$$\lambda = \frac{h}{p} \qquad \Delta x \cdot \Delta p \sim h$$

- If we precisely measure x , we do not know anything about p

Reviewing the double-slit experiment

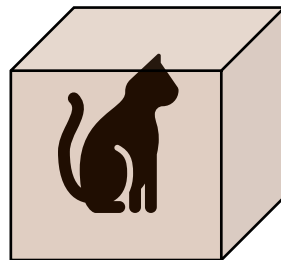


Schrödinger's cat



- Measurements affect the outcome – Schrödinger's cat

$$\psi(\mathbf{r}, t) = \frac{1}{\sqrt{2}}[\psi_{\text{dead}}(\mathbf{r}, t) + \psi_{\text{alive}}(\mathbf{r}, t)]$$



$$\psi(\mathbf{r}, t) = \psi_{\text{dead}}(\mathbf{r}, t)$$



$$\psi(\mathbf{r}, t) = \psi_{\text{alive}}(\mathbf{r}, t)$$